R Course
Assignment 1

Introduction

In this assignment, you will use basic R programming to simulate how insecticide resistance evolves in an insect population.

Consider a single gene with two alleles: S (susceptible to the insecticide, dominant) and r (resistant, recessive). Because r is recessive, only insects with genotype rr clearly survive spraying, insects with genotype SS or Sr still behave as susceptible. We describe this with a survival probability between 0 and 1 for each genotype, for example:

```
w_SS <- 0.35 # the probability of surviving to reproduce of genotype SS
w_Sr <- 0.35 # the probability of surviving to reproduce of genotype Sr
w_rr <- 1.0  # the probability of surviving to reproduce of genotype rr
```

Let q_n denote the relative frequency of the resistance allele r in generation n, and let $p_n = 1 - q_n$ be the relative frequency of S. Assuming random mating, the genotypes SS, Sr and rr occur in generation n with the Hardy-Weinberg proportions p_n^2, $2p_nq_n$ and q_n^2. After one round of selection by the insecticide, the frequency of the resistance allele in generation $n+1$ is

$$q_{n+1} = \frac{p_n q_n w_{Sr} + q_n^2 w_{rr}}{\bar{w}_n}, \qquad \bar{w}_n = p_n^2 w_{SS} + 2p_n q_n w_{Sr} + q_n^2 w_{rr},$$

where $\bar{w}_n$ is the mean survival probability in generation n.

Task

Create a new R script called `Assignment1_solution.R`.

1. Create a variable for the number of generations to simulate (25).

```
number_of_generations <- 25
```

2. Create variables `w_SS`, `w_Sr` and `w_rr` for the three genotype survival probabilities given above.

(As above)

3. Create a vector `generation` containing the numbers 0 to `number_of_generations`.

```
generations <- 0:number_of_generations
generations
```

```
##  [1]  0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
## [26] 25
```

4. Create a numeric vector `q` of length `number_of_generations + 1`, and set `q[1] <- 0.05` as the starting frequency of the resistance allele, before spraying began.

```
q <- rep(0, length.out = number_of_generations + 1)
q[1] <- 0.05
q
```

```
##  [1] 0.05 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
## [16] 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
```

5. Use a `for` loop to fill in the remaining entries of `q` based on the formulas above.

```
for ( n in 1:(length(q)-1) ) {
  q_n <- q[n]
  p_n <- 1 - q_n
  w_bar <- p_n * p_n * w_SS + 2 * p_n * q_n * w_Sr + q_n * q_n * w_rr
  q[n+1] <- (p_n * q_n * w_Sr + q_n * q_n * w_rr) / w_bar
}
q
```

```
##  [1] 0.05000000 0.05439033 0.05955712 0.06571164 0.07314426 0.08226277
##  [7] 0.09365337 0.10818014 0.12715069 0.15259403 0.18771983 0.23761301
## [13] 0.30996596 0.41444741 0.55606163 0.71799776 0.85592988 0.93896814
## [19] 0.97685883 0.99165237 0.99704641 0.99896227 0.99963630 0.99987265
## [25] 0.99995542 0.99998440
```

6. Modify the loop to also save the mean survival probability $\bar{w}_n$ in a vector `w_bar` of length `number_of_generations`.

```
w_bar <- rep(0, length.out = number_of_generations)
for ( n in 1:(length(q)-1) ) {
  q_n <- q[n]
  p_n <- 1 - q_n
  w_bar[n] <- p_n * p_n * w_SS + 2 * p_n * q_n * w_Sr + q_n * q_n * w_rr
  q[n+1] <- (p_n * q_n * w_Sr + q_n * q_n * w_rr) / w_bar[n]
}
q
```

```
##  [1] 0.05000000 0.05439033 0.05955712 0.06571164 0.07314426 0.08226277
##  [7] 0.09365337 0.10818014 0.12715069 0.15259403 0.18771983 0.23761301
## [13] 0.30996596 0.41444741 0.55606163 0.71799776 0.85592988 0.93896814
## [19] 0.97685883 0.99165237 0.99704641 0.99896227 0.99963630 0.99987265
## [25] 0.99995542 0.99998440
```

```
w_bar
```

```
##  [1] 0.3516250 0.3519229 0.3523056 0.3528067 0.3534776 0.3543987 0.3557011
##  [8] 0.3576069 0.3605087 0.3651352 0.3729052 0.3866990 0.4124513 0.4616483
## [15] 0.5509830 0.6850885 0.8262004 0.9230798 0.9702646 0.9891934 0.9961660
## [22] 0.9986516 0.9995273 0.9998344 0.9999420
```

7. To see how the strength of the insecticide affects the speed of resistance evolution, repeat the simulation for a weaker insecticide (`w_SS <- 0.6` and `w_Sr <- 0.6`). Use a `for` loop, as before, to compute this second frequency trajectory and store it in a new vector `q_weak`.

```
w_SS_weak <- 0.6
w_Sr_weak <- 0.6
q_weak <- rep(0, length.out = number_of_generations + 1)
q_weak[1] <- 0.05
w_bar_weak <- rep(0, length.out = number_of_generations)
for ( n in 1:(length(q)-1) ) {
  q_n <- q_weak[n]
  p_n <- 1 - q_n
  w_bar_weak[n] <- p_n * p_n * w_SS_weak + 2 * p_n * q_n * w_Sr_weak + q_n * q_n * w_rr
  q_weak[n+1] <- (p_n * q_n * w_Sr_weak + q_n * q_n * w_rr) / w_bar_weak[n]
}
q_weak
```

```
##  [1] 0.05000000 0.05158070 0.05325994 0.05504693 0.05695199 0.05898679
##  [7] 0.06116454 0.06350023 0.06601097 0.06871631 0.07163875 0.07480422
## [13] 0.07824279 0.08198946 0.08608518 0.09057812 0.09552524 0.10099423
## [19] 0.10706608 0.11383822 0.12142857 0.12998080 0.13967098 0.15071619
## [25] 0.16338552 0.17801404
```

```
w_bar_weak
```

```
##  [1] 0.6010000 0.6010642 0.6011346 0.6012121 0.6012974 0.6013918 0.6014964
##  [8] 0.6016129 0.6017430 0.6018888 0.6020528 0.6022383 0.6024488 0.6026889
## [15] 0.6029643 0.6032818 0.6036500 0.6040799 0.6045853 0.6051837 0.6058980
## [22] 0.6067580 0.6078032 0.6090861 0.6106779
```

8. To see how the starting frequency of the resistance allele interacts with the strength of selection, repeat the simulation once more using a lower starting frequency `q[1] <- 0.01` combined with a much stronger insecticide (`w_SS <- 0.1` and `w_Sr <- 0.1`). Use a `for` loop, as before, to compute this trajectory and store it in a new vector `q_lowstart`. Even though it starts out behind the other two scenarios, the stronger selection should let it overtake both of them at some point during the simulation.

```
w_SS_lowstart <- 0.1
w_Sr_lowstart <- 0.1
q_lowstart <- rep(0, length.out = number_of_generations + 1)
q_lowstart[1] <- 0.01
w_bar_lowstart <- rep(0, length.out = number_of_generations)
for ( n in 1:(length(q)-1) ) {
  q_n <- q_lowstart[n]
  p_n <- 1 - q_n
  w_bar_lowstart[n] <- p_n * p_n * w_SS_lowstart + 2 * p_n * q_n * w_Sr_lowstart + q_n * q_n * w_rr
  q_lowstart[n+1] <- (p_n * q_n * w_Sr_lowstart + q_n * q_n * w_rr) / w_bar_lowstart[n]
}
q_lowstart
```

```
##  [1] 0.01000000 0.01089020 0.01194482 0.01321196 0.01475977 0.01668771
##  [7] 0.01914605 0.02237138 0.02675517 0.03298522 0.04236262 0.05758388
## [13] 0.08489353 0.14063391 0.27048785 0.56013024 0.88496266 0.98570686
## [19] 0.99853322 0.99985293 0.99998529 0.99999853 0.99999985 0.99999999
## [25] 1.00000000 1.00000000
```

```
w_bar_lowstart
```

```
##  [1] 0.1000900 0.1001067 0.1001284 0.1001571 0.1001961 0.1002506 0.1003299
##  [8] 0.1004504 0.1006443 0.1009792 0.1016151 0.1029843 0.1064862 0.1178001
## [15] 0.1658473 0.3823713 0.8048430 0.9744562 0.9973617 0.9997353 0.9999735
## [22] 0.9999974 0.9999997 1.0000000 1.0000000
```

9. Use a `while` loop to find the first generation in which `q_lowstart` overtakes `q`, and print that generation.

```
index <- 1
while (q_lowstart[index] <= q[index]) {
  index <- index + 1
}
paste("Generation:", index)
```

```
## [1] "Generation: 17"
```

10. Create a plot of `q`, `q_weak` and `q_lowstart` against `generation`. Add a title, labels for both axes and a legend distinguishing the three scenarios.

```
plot(generations, q, type = "l", col = "#002060", lwd = 2, ylim = c(0, 1),
     main = "Spread of allele r",
     xlab = "Generation", ylab = "Frequency of allele r")
lines(generations, q_weak, col = "blue", lwd = 2)
lines(generations, q_lowstart, col = "#00B0F0", lwd = 2)
legend(
  "topleft",
  legend = c("q", "q_weak", "q_lowstart"),
  col = c("#002060", "blue", "#00B0F0"), lwd = 2, cex = 0.8, bg = "white"
)
```

11. Add a short comment explaining what the simulation suggests about the effect of insecticide strength and starting frequency on how quickly resistance spreads.

> *The simulation suggests that the starting frequency does not matter for the spread of the allele. The strength of the insecticide determines the rate at wich the resistance spreads.*

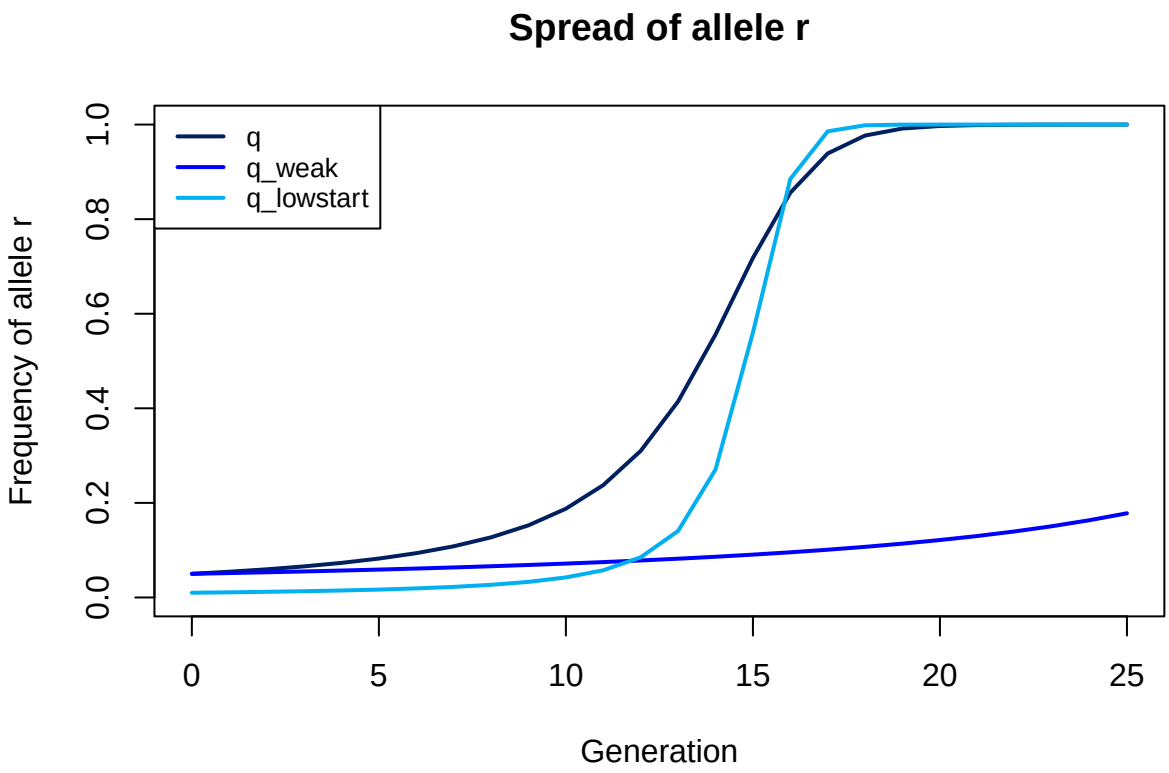


Abbildung 1: Frequency of the resistance allele r over the generations for the baseline scenario (q), a weaker insecticide (q_weak) and a lower starting frequency combined with a stronger insecticide (q_lowstart).